From Seasonal Stability to Event-Driven Dynamics: A Spectral Analysis of Groundwater Response to Winter Conditions Near JFK Airport, New York

Mighty, M.[1], Neville, J.[1], Bernhardt, L.[1], Tsakiri, K.G.[2], Marsellos, A.E.[1]

[1]*Department of Geology, Environment and Sustainability, Hofstra University, Hempstead, NY 11549*
[2]*Dept. Information Systems, Analytics, and Supply Chain Management, Rider University,* Lawrenceville, NJ 08648

Abstract

Long Island depends on a designated sole-source aquifer system, and this groundwater resource is increasingly vulnerable to seasonal road-salt application. This study focuses on the highly urbanized corridor surrounding John F. Kennedy International Airport (JFK), where winter maintenance and impervious surfaces can generate pulses of chloride-rich meltwater during snow events. Rather than directly measuring salt application, this study uses winter temperature conditions as a proxy for likely salting activity. To evaluate how these seasonal conditions relate to groundwater behavior, we analyzed long-term USGS groundwater-level records from wells near JFK. Winters were classified based on the frequency of near-freezing events (−4°C to 1°C), with winters exhibiting one or more such events defined as maximum-salt conditions, and winters with no such events and predominantly warmer temperatures defined as minimum-salt conditions. Spatial and time-series methods, including spectral analysis, periodograms, and wavelet analysis, were used to examine seasonal and multi-year variability. Results show that winters with more frequent near-freezing conditions exhibit reduced low-frequency variability and increased short-term fluctuations, indicating a shift toward a more dynamic and less buffered aquifer response. These findings suggest that winter environmental conditions may alter groundwater system behavior in urban settings. Spectral methods were applied to detect seasonal forcing, multi-year oscillations, and evolving recharge dynamics that cannot be identified solely through linear trend analysis. These methods are used to identify patterns in groundwater response rather than to directly demonstrate the effects of road salt. Differences in spectral power and periodic structure between winter groups are interpreted as potential changes in aquifer response under differing winter conditions. Although the magnitude and specific mechanisms of these effects remain uncertain, the results provide insight into how winter conditions may influence groundwater behavior in urban environments. This work highlights the potential impact of winter conditions on aquifer health and underscores the importance of considering winter maintenance practices in long-term groundwater management.

Introduction

Groundwater is a critical resource for Long Island, which relies on a sole-source aquifer, system for drinking water and ecosystem sustainability (United States Environmental Protection Agency, 2025; New York State Department of Transportation, 2018). In highly urbanized environments such as the region surrounding John F. Kennedy International Airport (JFK), groundwater systems are increasingly influenced by anthropogenic processes, including impervious surface development, stormwater routing, and winter road maintenance practices. Among these, the application of road salt during freeze–thaw conditions is a major driver of freshwater salinization and altered hydrologic behavior (Schlesinger & Bernhardt, 2019; Rodbarry et al., 2023).

Winter conditions introduce highly variable forcing mechanisms to groundwater systems. Freeze–thaw cycles and snowmelt events generate pulses of runoff that infiltrate recharge zones and modify both the chemical and physical behavior of aquifers. While long-term increases in chloride concentrations have been widely documented (Rodbarry et al., 2023; United States Environmental Protection Agency, 2025), less attention has been given to how these seasonal processes influence the temporal structure of groundwater variability across multiple time scales. Unlike many pollutants, chloride behaves conservatively in aquatic systems, meaning it does not readily degrade or transform. As a result, salt introduced during winter months can persist year-round, contributing to sustained increases in background salinity levels (Stroud Water Research Center, 2026; Schlesinger & Bernhardt, 2019).

Long Island's aquifer system is particularly vulnerable to this type of contamination due to its high permeability, shallow water table, and dependence on local recharge (New York State Department of Transportation, 2018). These characteristics allow surface-derived contaminants to infiltrate rapidly into groundwater. In urban environments, extensive impervious surfaces and stormwater infrastructure further enhance this process by concentrating and redirecting runoff toward recharge areas. Chloride from road salt has been shown to infiltrate drinking water supplies, degrade aquatic ecosystems, and contribute to infrastructure corrosion and mobilization of toxic metals (Rodbarry et al., 2023; Schlesinger & Bernhardt, 2019).

Recent regional assessments highlight the growing scale of this issue. The New York City Department of Environmental Protection (DEP) has documented steadily increasing chloride concentrations across all major water supply systems since the 1980s, with the most rapid increases occurring in the more urbanized East of Hudson watershed (Bureau of Water Supply, 2025). Projections suggest that continued trends could eventually approach or exceed recommended drinking water thresholds, emphasizing the need for proactive management. Although these studies primarily focus on surface reservoirs, they reflect a broader pattern of freshwater salinization occurring across developed regions (Schlesinger & Bernhardt, 2019).

Traditional hydrologic analyses often rely on trend-based or statistical approaches that may overlook non-stationary and multi-scale behavior. However, groundwater systems are inherently dynamic, with variability driven by interacting processes operating at seasonal, interannual, and short-term scales. Previous work has demonstrated that time series decomposition and signal detection techniques are essential for separating these components and reducing noise in environmental datasets (Parag et al., 2025). Spectral methods such as periodograms and wavelet analysis provide a framework for identifying dominant periodicities and time-frequency variability that cannot be captured through linear analysis alone (Parag et al., 2025).

To address this gap, this study examines groundwater-level responses to winter conditions near JFK Airport using time-series and spectral analysis techniques. Winter conditions are classified using temperature-based proxies for freeze–thaw activity associated with increased road salt application. By comparing groundwater variability between winters with minimal and more frequent freeze conditions, this study aims to determine whether seasonal processes are associated with measurable changes in aquifer response patterns. These findings provide insight into how winter environmental conditions may influence groundwater dynamics in highly urbanized regions and contribute to the broader understanding of freshwater salinization.

Methodology

Groundwater-level and chloride data were retrieved from the U.S. Geological Survey (USGS) Water Data Services using the R package dataRetrieval (De Cicco et al., 2023). Chloride concentrations (parameter code 00940, mg/L) and daily groundwater levels (parameter code 62610, depth to water) for 2000–2024 were obtained for wells located within 10 km of John F. Kennedy International Airport (JFK; −73.7781, 40.6413). The JFK groundwater is a part of both the Jameco and Upper Glacial aquifers. Groundwater wells were first identified within a bounding box around JFK using the U.S. Geological Survey’s National Water Information System (NWIS) and then clipped to a 10 km buffer. Spatial analyses were conducted in NAD83 / New York Long Island State Plane (EPSG: 2263), and the 10 km buffer was converted from meters to U.S. survey feet to match the projection units. Wells were retained if their coordinates fell within the buffer, and records lacking geographic coordinates were excluded.

Figure 1: Groundwater level time series (2000–2024) from the well located closest to the highest chloride concentration near JFK Airport, with data quality control applied. Observed data are shown in black, short gaps (≤14 consecutive days) filled using linear interpolation in blue, and longer gaps left unfilled (red markers). This approach preserves short-term variability while avoiding artificial smoothing that could bias spectral and wavelet analyses.

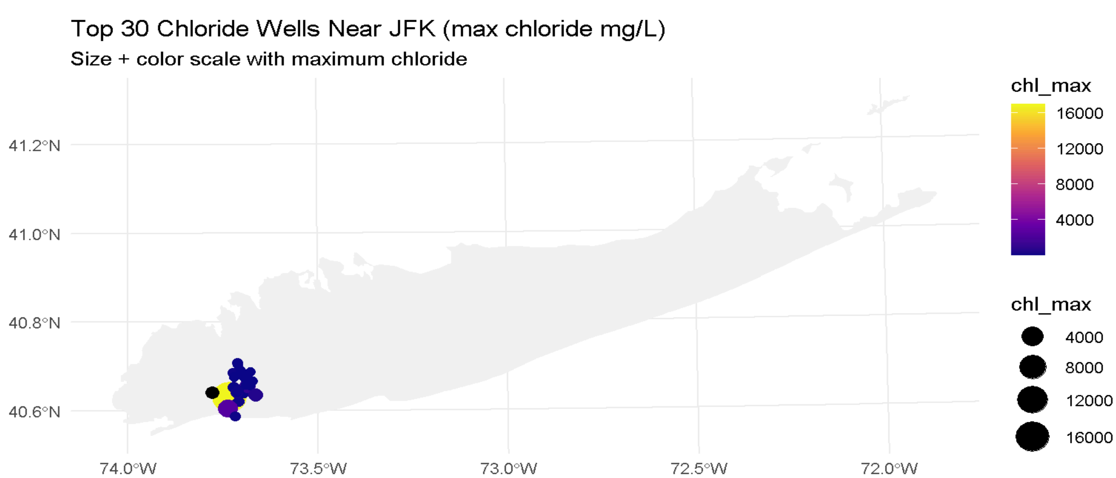

Figure 2: Spatial distribution of the 30 groundwater wells with the highest recorded chloride concentrations near JFK Airport (within 10km radius). Point size and color represent maximum observed chloride concentration (mg/L), with larger and warmer-colored symbols indicating higher values. Wells are clustered within the highly urbanized coastal zone, highlighting areas of elevated salinity and guiding selection of the representative groundwater time series used for spectral analysis.

Groundwater-level records were screened for continuity, as shown in Figure 1. Chloride metadata were used to identify wells with adequate sampling histories, as shown in Figure 2. The groundwater-level well with the longest continuous daily record was selected for detailed spectral analysis. Duplicate dates were removed, and short gaps (≤14 days) were linearly interpolated; longer gaps were left unfilled to avoid spectral distortion. To preserve temporal continuity while minimizing distortion of spectral properties, short gaps (≤14 consecutive days) were interpolated using a linear method implemented via the imputeTS package (na_interpolation). This threshold was selected to avoid introducing artificial periodicities, as interpolation across longer gaps can impose smooth transitions that generate spurious low-frequency signals. By restricting interpolation to short intervals, the procedure maintains the integrity of high-frequency variability while preventing bias in spectral and wavelet analyses. December observations were assigned to the following hydrologic winter year. Winters were defined as December–February (DJF) and classified using a temperature-based proxy for likely road-salt application. Daily modeled temperature data (2000–2024) were used to identify near-freezing events (−4°C to 1°C) indicative of freeze/snow maintenance conditions (National Weather Service, n.d.). Near-freezing conditions are strongly associated with snow and ice formation, which typically trigger road salt application in urban environments. Winters with no such events were classified as minimum-salt (min-salt), while winters with one or more events were classified as maximum-salt (max-salt). This classification represents a proxy for relative salting intensity rather than a direct measure of salt application. Daily precipitation data for JFK were obtained from NOAA's Global Historical Climatology Network (GHCN) via the R package GHCNr (National Centers for Environmental Information, n.d.) to provide seasonal context. Precipitation time series were additionally smoothed using moving averages and Kolmogorov–Zurbenko (KZ) filtering to examine seasonal variability. Groundwater data were subset to DJF and analyzed using periodograms and wavelet analysis (Hernandez Gonzalez et al., 2023; Parag et al., 2025). Spectral methods were applied as exploratory tools to identify periodic structure, seasonal forcing, and time-frequency variability in groundwater response, rather than to demonstrate direct causal effects. The DJF series was evaluated both as concatenated seasonal records (exploratory) and as per-winter-averaged power spectral densities to reduce bias arising from artificial adjacency between winters. Comparisons between min-salt and max-salt winters focused on differences in spectral power, dominant periodicities, and frequency structure. Variations in these properties were interpreted as changes in aquifer response under varying winter conditions, with increased spectral power indicating a stronger or more dynamic groundwater response. These analyses evaluate whether winters with more frequent freeze-and-snow maintenance conditions are associated with changes in groundwater behavior near JFK, rather than establishing road salt as the direct cause. Analyses were conducted in R (version 4.5.3) using the following packages: dataRetrieval, GHCNr, ggplot2, ggspatial, kza, lubridate, purrr, sf, tigris, tidyverse, trend, USAboundaries, viridis, WaveletComp, and zoo. Session details and download dates were saved, and results reflect the USGS data available at the time of retrieval.

Temperature-Based Classification of Winter Salt Regimes

Daily air temperature data from NOAA's *Global Surface Summary of the Day (GSOD)* were used to classify winters based on thermal conditions associated with road-salt application. Winters were categorized as maximum-salt when at least one near-freezing interval (−4°C to 1°C) occurred, and as minimum-salt when such intervals were absent and temperatures were predominantly within a warmer range (+4°C to +10°C). This classification follows the temperature-based proxy framework introduced by Gonzalez et al. (2023) for assessing climate-driven processes affecting geomorphological systems, including sinkhole formation.

Methodological Considerations: Seasonal Segmentation and Spectral Integrity

A key methodological consideration in this study is the treatment of discontinuous seasonal data in spectral and wavelet analyses. When DJF segments from multiple years are concatenated into a single time series, artificial adjacency is introduced between the end of one winter and the beginning of the next. This creates discontinuities that do not represent real hydrologic processes and can introduce broadband spectral energy, potentially distorting periodograms, wavelet power, and the interpretation of dominant timescales. To avoid these effects, winters are analyzed independently, preventing the mixing of temporally unrelated observations. In the present analysis, concatenated DJF series were used only for exploratory visualization, while the primary results rely on per-winter spectral estimation and subsequent averaging across winters. By computing multitaper spectra and wavelet power independently for each winter and then aggregating these estimates, we reduce bias associated with artificial boundary effects and preserve the intrinsic temporal structure of each season. This approach ensures that detected periodicities reflect within-winter groundwater dynamics rather than artifacts of data construction, thereby strengthening the robustness of the inferred differences between minimum-salt and maximum-salt winter conditions.

Results

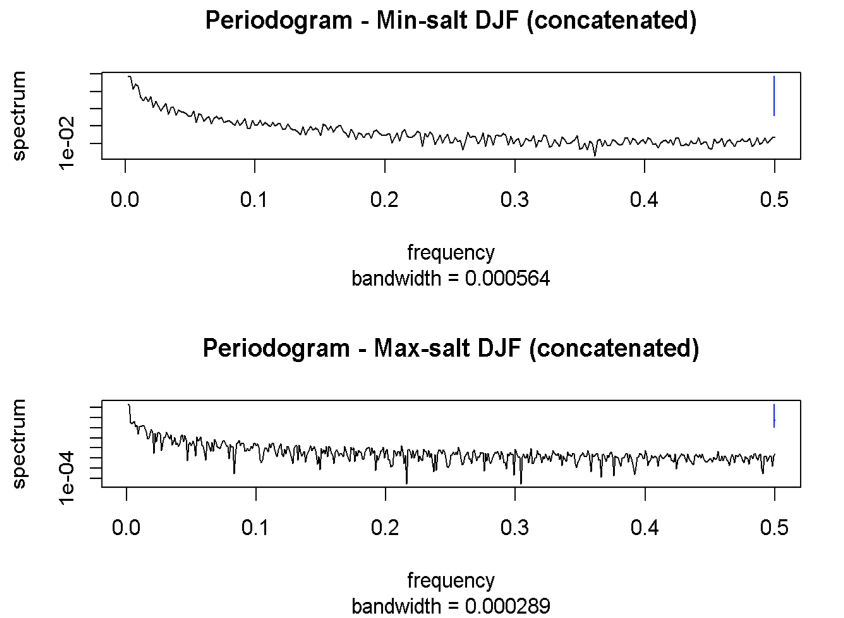

Figure 3. Periodogram analysis of groundwater levels during DJF periods showing differences in frequency structure between minimum-salt and maximum-salt winters.

Periodogram analysis of groundwater levels during DJF periods reveals a clear difference in frequency structure between minimum-salt and maximum-salt winters. As shown in Figure 3, minimum-salt winters exhibit spectral energy concentrated at low frequencies, indicating dominance of seasonal-scale variability and a relatively smooth, well-filtered groundwater response. In contrast, maximum-salt winters exhibit reduced overall spectral power and a more irregular distribution of energy across frequencies, including increased variability in the mid- and high-frequency ranges. This pattern suggests a shift from a stable, low-frequency-dominated system to a more complex, less filtered response under presumed high-salt conditions. The reduction in spectral coherence and increase in higher-frequency variability indicate that groundwater dynamics become more irregular during winters associated with increased freeze–thaw activity and snow maintenance practices, conditions associated with increased road salt application.

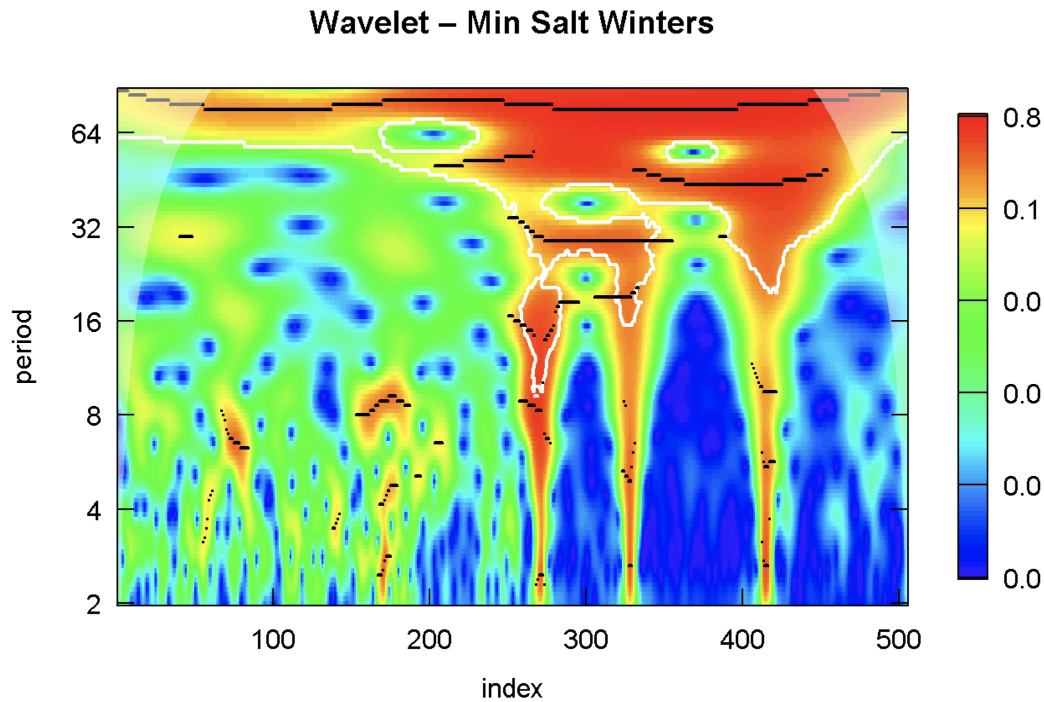

Figure 4. Wavelet power spectrum of groundwater levels during minimum-salt winters, showing dominant spectral energy at longer periods (~30–90 days) and a continuous band indicative of stable, seasonally driven variability.

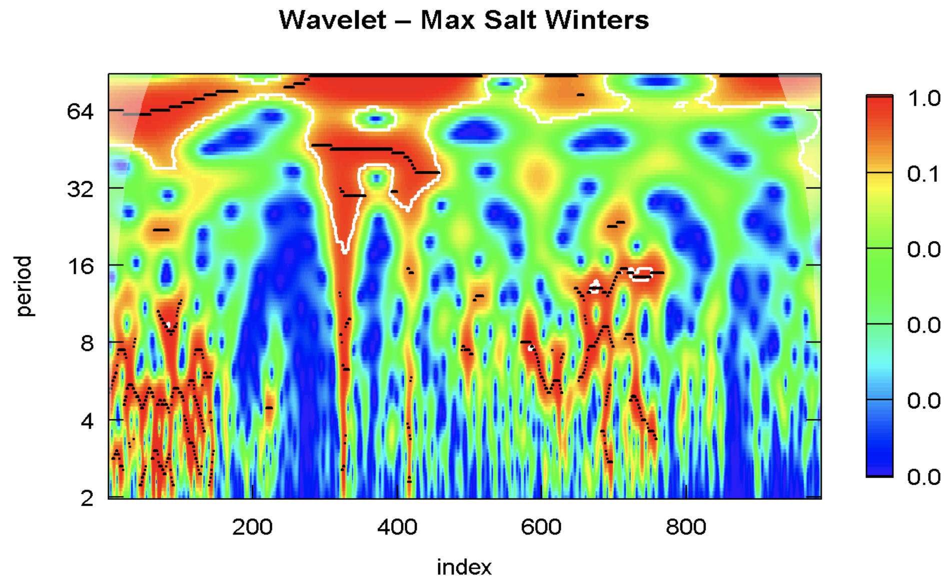

Figure 5. Wavelet power spectrum of groundwater levels during maximum-salt winters, showing a broader and more fragmented distribution of spectral power across both long and short periods, indicating increased short-term variability and reduced temporal coherence.

As shown in Figures 4 and 5, the color gradients represent wavelet plots of spectral power as a function of time and period, with warmer colors (red and orange) indicating higher power and cooler colors (blue) indicating lower power. In the minimum-salt winters, high-power regions are concentrated in the upper portion of the plots (longer periods), forming continuous bands that reflect persistent seasonal signals. In contrast, maximum-salt winters exhibit high-power regions that are more widely distributed across both long and short periods, with frequent instances of elevated power at shorter timescales. This broader and more irregular distribution indicates that variability is both stronger and less temporally consistent. The presence of intermittent high-power regions at short periods further suggests increased influence of short-term fluctuations, reinforcing the interpretation that aquifer response becomes more variable and less filtered under high-salt conditions.

Wavelet analysis reveals clear differences in the temporal structure of groundwater variability between minimum-salt and maximum-salt winters. Minimum-salt winters exhibit a strong, continuous band of spectral power at longer periods (~30–90 days), indicating the dominance of stable, seasonal-scale groundwater fluctuations. This pattern remains consistent through time, reflecting a relatively coherent and predictable recharge regime. In contrast, maximum-salt winters display a more fragmented distribution of spectral power, with increased energy at shorter periods (2–16 days) and reduced continuity of the seasonal band. These patterns indicate that groundwater response becomes less stable and more irregular under high-salt conditions, reflecting a shift from a consistent, seasonally driven system toward a more dynamically variable and less filtered aquifer response during winters associated with increased freeze–thaw activity.

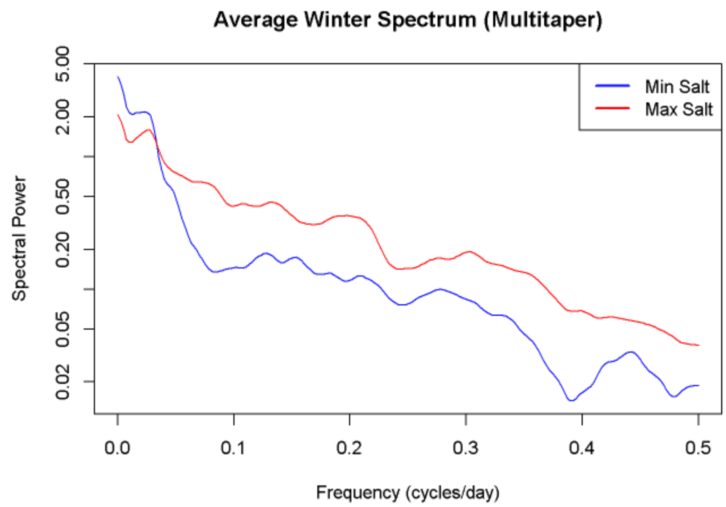

Figure 6. Frequency-domain spectral comparison of groundwater levels during minimum-salt and maximum-salt winters, showing reduced low-frequency power and a flatter spectral distribution under maximum-salt conditions.

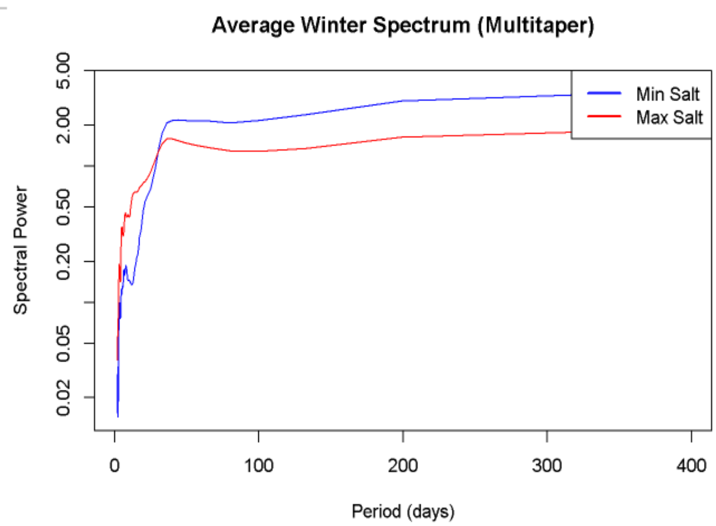

Figure 7. Period-domain spectral comparison of groundwater levels, illustrating reduced spectral power at longer periods (~100–300 days) during maximum-salt winters, indicating diminished seasonal variability.

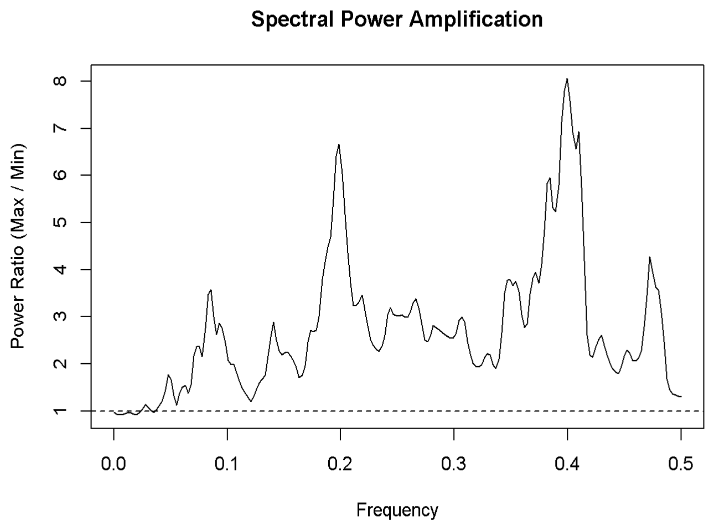

Figure 8. Spectral power ratio (maximum-salt divided by minimum-salt) across frequencies, showing amplification of groundwater variability under maximum-salt conditions, particularly at higher frequencies.

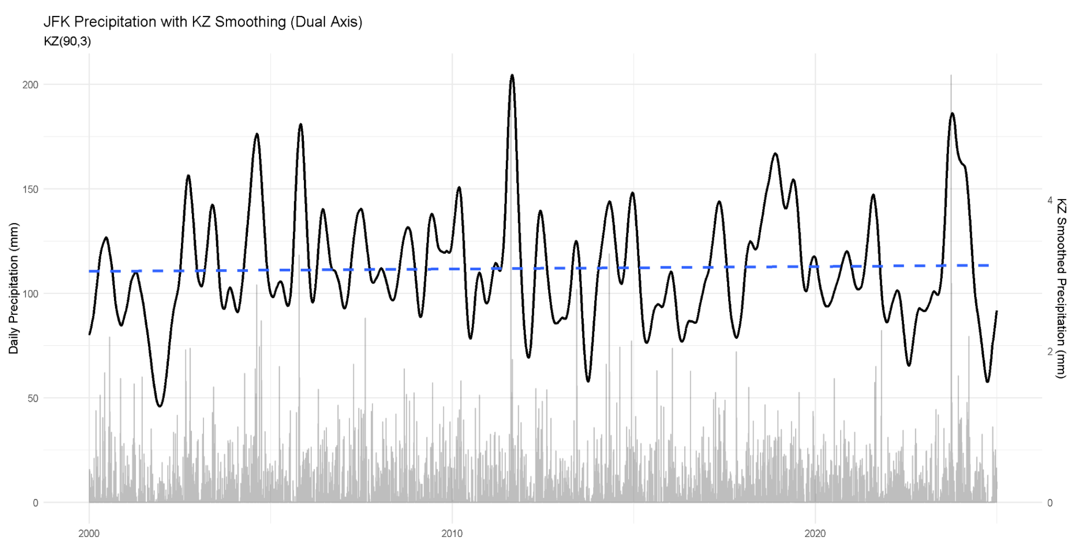

Figure 9. Daily precipitation at JFK (2000–2024) shown as gray bars, with a Kolmogorov–Zurbenko (KZ) smoothed series ($m = 90$, $k = 3$) overlaid as a black line (right axis).

As shown in Figures 7 and 8, spectral comparisons in both the period and frequency domains further highlight differences between minimum-salt and maximum-salt conditions. The frequency-domain spectral comparison shows clear differences in groundwater variability between minimum-salt and maximum-salt winters. Minimum-salt winters exhibit higher spectral power at low frequencies, indicating a strong dominance of long-timescale (seasonal) groundwater fluctuations. In contrast, maximum-salt winters display reduced power in this low-frequency range and a comparatively flatter spectral distribution, suggesting that the seasonal signal is less pronounced under high-salt conditions. At higher frequencies, representing short-term variability, the differences between the two groups are smaller, indicating that short-period fluctuations remain present in both cases. Overall, the reduced low-frequency power in maximum-salt winters indicates a weakening of structured, seasonal groundwater dynamics.

The period-domain spectral comparison further highlights the attenuation of seasonal groundwater variability during maximum-salt winters. Minimum-salt winters show greater spectral power at longer periods (~100–300 days), corresponding to strong and coherent seasonal recharge signals. In contrast, maximum-salt winters exhibit consistently lower spectral power across these longer periods, indicating a diminished seasonal component. Differences between the two groups are less pronounced at shorter periods, suggesting that short-term variability is relatively similar between conditions. This pattern confirms that the primary distinction between winter types lies in the strength of long-period variability, with maximum-salt winters characterized by a reduction in dominant seasonal groundwater behavior.

To assess statistical significance, a Welch two-sample t-test was conducted to compare spectral power between minimum-salt and maximum-salt winters, using log-transformed spectral estimates. The results indicate a statistically significant difference between the two groups ($t = -5.70$, $df = 383.62$, $p < 0.001$). The average spectral power for maximum-salt winters was higher than for minimum-salt winters, particularly at higher frequencies, with the 95% confidence interval for the difference ranging from -1.21 to -0.59, indicating consistently greater variability under presumed higher-salt conditions. These results confirm that the observed differences in spectral structure are not due to random variation, but instead reflect a systematic shift in groundwater dynamics associated with winter conditions characterized by increased freeze–thaw activity. The statistical significance of this result supports the interpretation that groundwater systems exhibit enhanced and more variable responses during maximum-salt winters.

The spectral power ratio (maximum-salt divided by minimum-salt) shown in Figure 8 shows consistent amplification of groundwater variability under presumed high-salt conditions across the entire frequency spectrum. Values greater than 1 across nearly all frequencies indicate that maximum-salt winters exhibit greater variability than minimum-salt winters at both seasonal and short timescales. Notably, amplification is particularly pronounced at higher frequencies, where power ratios range from 5 to 8, indicating strong enhancement of short-period fluctuations. This pattern demonstrates that the influence of winter conditions associated with frequent freeze–thaw events is not limited to a specific frequency band but instead reflects a broad increase in groundwater variability. These findings reinforce previous spectral and wavelet analyses, confirming that groundwater systems under high-salt conditions exhibit more dynamic and less filtered behavior.

As shown in Figure 9, analysis of daily precipitation at JFK indicates no significant long-term trend in precipitation, despite short-term variability in precipitation, providing hydrologic context for groundwater variability analyses, characterized by frequent fluctuations and intermittent high-magnitude events. Notably, extreme precipitation events, such as those associated with Hurricane Sandy, appear as pronounced peaks in the record, highlighting the variable nature of intense rainfall in the region. To isolate longer-term patterns from this high-frequency noise, a Kolmogorov–Zurbenko (KZ) filter (90-day window, 3 iterations) was applied. This smoothing technique effectively removes short-term variability while preserving seasonal-to-interannual structure, allowing for clearer interpretation of underlying trends. The resulting smoothed precipitation series remains relatively stable over the study period, fluctuating around a consistent mean without exhibiting a sustained directional trend. Despite the presence of extreme events, overall precipitation appears to be mean-reverting. The combination of high short-term variability and long-term stability suggests that observed changes in groundwater dynamics are unlikely to be driven by changes in precipitation input, but instead reflect variations in aquifer response behavior.

Discussion

The results of this study indicate that groundwater behavior near JFK Airport differs between winters with minimal freeze–thaw activity and those with more frequent events. Spectral and wavelet analyses show that winters with more frequent near-freezing events exhibit greater variability across seasonal and shorter timescales, suggesting a shift toward a more dynamic, less filtered groundwater system. In contrast, winters with fewer freeze events tend to exhibit more stable, low-frequency-dominated behavior consistent with a buffered aquifer response. These patterns are consistent with previous research showing that winter processes, including those associated with road salt application, can influence groundwater systems by increasing chloride inputs and altering subsurface flow dynamics. However, because this study uses temperature as a proxy rather than direct salt measurements, the results should be interpreted as reflecting broader winter environmental effects rather than road salt application alone. Other factors, such as freeze–thaw impacts on soil structure, precipitation variability, and urban runoff, may also contribute to the observed patterns. Additionally, precipitation was considered but not emphasized in the main analysis for clarity. Precipitation is an important factor influencing both road salt application and groundwater recharge, and may therefore affect the study in multiple ways. However, precipitation data in Figure 9 show no significant long-term variability in precipitation over the course of a 24-year period sufficient to influence aquifer recharge, as indicated by the relatively stable long-term average. Barring some exceptions, precipitation data indicates peaks balanced by corresponding low periods that average out over time, reducing its influence on the observed groundwater variability. Due to precipitation having an adverse effect on the amount of salt applied during maximum salt times, precipitation was still recorded and specific periods were analyzed, but found to have a negligible influence on the results. Cross-referencing temperature and precipitation could introduce additional complexity in a study that already uses temperature as a proxy, with the expectation that there are several external factors affecting aquifer recharge conditions. Overall, these findings suggest that winter conditions play an important role in shaping groundwater variability in urban coastal aquifers and highlight the need to consider seasonal processes in groundwater management.

Study Area: JFK as a Road Salt Case Study

This study focuses on a localized area surrounding JFK in Jamaica, Queens, rather than the broader NYC metropolitan region. New York City derives approximately 90% of its potable water from major reservoir systems; the latter half consisting of lakes, rivers, The New Croton Reservoir and Queen's groundwater supply system in the southeastern region subsisting the 10%. The Catskills and Delaware watersheds make a majority of NYC's drinking water through reservoir collection and distribution (National Center for Biotechnology Information, 2021). The Croton watershed, located north of New York City and separate from the Long Island aquifer system, has experienced salinity increases due to geological factors, local wastewater treatment plants, fractured bedrock and glacial soils, slow dispersal of contamination, and as well as suburban activity, which may limit its suitability for isolating urban-scale effects compared to JFK. In 2021, the New York City Department of Environmental Protection had tasked an internal Salinity Task Force in order to better understand salinity in the NYC watersheds through its analysis stemming as far as 1985 in upstate's reservoirs. Following their investigation the task force found a sustained increase in chloride concentrations in all NYC's reservoirs. The West of Hudson watersheds have relatively low levels of chloride, compared to the East of Hudson's watersheds, with the highest increases recorded in The Croton watershed (NYC Department of Environmental Protection, 2025). While the Jamaica, Queens groundwater system contributes less than 10% of NYC's potable water, the suburban and city implications to salt application represent a far larger trend followed by cities in snowbelts. JFK's local wells and aquifers concisely isolate numerous factors plaguing larger regions of study regarding road salt's increased freshwater salinity, including anthropogenic weathering, wastewater inputs, urban construction, fertilizer use, resource extraction, water softening, saltwater intrusion, and climate change. While JFK is not limited to road salt intrusion and subjected to salinity increases via external factors, it provides a unique advantage as a study area due to its concentrated and consistent winter salt application within its large surface area characterized by intensive salt application during winter conditions.

Interpretation of Spectral Power and Timescales in Groundwater Dynamics

Spectral analysis helps interpret groundwater dynamics across multiple timescales by decomposing variability into frequency components. In this context, low frequencies (long periods) represent seasonal to multi-month groundwater behavior associated with recharge, storage, and delayed release processes, while high frequencies (short periods) reflect rapid responses to individual events such as precipitation pulses, snowmelt, or transient surface inputs.

Spectral power (or spectral density) quantifies the strength of variability at each timescale. Higher spectral power indicates that fluctuations at a given period are more pronounced and consistent, meaning that the corresponding process strongly influences groundwater behavior. In hydrologic terms, strong low-frequency power reflects a coherent and efficient aquifer system, where recharge is stored and released gradually, producing stable seasonal signals. In contrast, elevated high-frequency power indicates a system that responds rapidly to short-term forcing, suggesting reduced buffering capacity and more event-driven dynamics.

In this study, minimum-salt winters exhibit stronger spectral power at longer periods, indicating a dominant seasonal recharge signal and a more stable, storage-controlled aquifer response. Conversely, maximum-salt winters show increased power at shorter periods and reduced coherence at longer periods, reflecting a shift toward more variable and less filtered groundwater dynamics. This redistribution of spectral energy suggests that aquifer behavior transitions from a regulated, storage-dominated system to a more reactive system influenced by short-term inputs under salt-intensive winter conditions.

It is important to note that a transition in behavior is observed around ~30–40 days, representing a characteristic timescale separating short-term event-driven responses from longer-term storage processes. This crossover provides insight into the effective "memory" of the aquifer system, beyond which groundwater dynamics become dominated by seasonal recharge rather than transient inputs. This transition is consistent with the observed crossover in spectral power between minimum-salt and maximum-salt winters, where short-period variability becomes dominant under high-salt conditions while long-period structure weakens (Figure 7).

Conclusion

This study examined how winter conditions associated with freeze–thaw activity influence groundwater behavior near JFK Airport using spectral, time–frequency, and statistical analyses. Across multiple methods, consistent differences were observed between minimum-salt and maximum-salt winters, indicating that winter conditions are associated with measurable changes in aquifer response dynamics. Spectral analyses, including periodograms and multitaper spectral estimates, show that minimum-salt winters are characterized by stronger low-frequency (seasonal-scale) groundwater variability, reflecting a stable and well-structured recharge regime. In contrast, maximum-salt winters exhibit reduced spectral power at these low frequencies and a comparatively flatter spectral distribution. This indicates a weakening of the dominant seasonal groundwater signal rather than a simple increase in overall variability. Period-domain representations further support this interpretation, showing diminished power at longer periods (~90 days) during maximum-salt winters, consistent with attenuation of coherent seasonal recharge processes. Wavelet analysis reinforces these findings by demonstrating that minimum-salt winters maintain continuous, high-power bands at longer periods, indicative of stable seasonal forcing. In contrast, maximum-salt winters display more fragmented and temporally inconsistent patterns, with intermittent high-power regions across both long and short timescales. This suggests that groundwater dynamics under these conditions become less temporally coherent and more irregular.

Statistical analysis confirms that these differences are significant. The Welch two-sample t-test indicates a clear separation in spectral characteristics between winter types, demonstrating that the observed changes in groundwater variability are systematic rather than random. Additionally, spectral power ratio analysis shows that while short-term variability may be amplified in maximum-salt winters, this occurs alongside a reduction in structured seasonal behavior, indicating a shift in how variability is distributed across timescales rather than a uniform increase in system energy. Importantly, precipitation analysis provides critical context for interpreting these results. Despite substantial short-term variability and episodic extreme events, including large precipitation peaks associated with storms such as Hurricane Sandy, the long-term smoothed precipitation record (via KZ filtering) remains stable and does not exhibit a sustained trend. This indicates that precipitation has remained relatively constant over the study period. As a result, the observed changes in groundwater behavior are unlikely to be driven directly by changes in rainfall input, but instead reflect alterations in aquifer response dynamics.

Overall, the consistency across multiple analytical approaches (periodograms, wavelet transforms, spectral comparisons, and statistical testing) indicates that the observed differences are robust. These findings suggest that winter conditions associated with frequent freeze–thaw activity are linked to measurable changes in groundwater behavior, characterized by increased variability and altered frequency structure. While this study does not directly confirm these changes to road salt application, the results are consistent with the hypothesis that environmental conditions associated with salt-intensive winters (such as sodium chloride application) may influence aquifer response dynamics in the JFK Upper Glacial and Jameco aquifers. A suggestion for future research would be to access data of actual existing NaCl road salt application (instead of proxy data) times in the Jameco and Upper glacial aquifer of the target region.

Credit Authorship Contribution Statement

Neville, J: Editing, writing- Results, Conclusion, Probatory R-studio Coding-figure 7, unused CCF lag graphs, research groundwork, R-Studio reviewing; Marsellos, A.E.: supervision, Rstudio coding, guidance, editing, Discussion; Tsakiri, K.G.: Rstudio coding review; Mighty, M: peer researcher, research groundwork, writing- discussion, editing; Bernhardt, L: writing - abstract and introduction, research, Rstudio coding, revising.

Acknowledgements

This study would not have been possible without Dr. Antonios E. Marsellos and his Geoinformatics Laboratory in the Department of Geology, Environment and Sustainability at Hofstra University for providing the computational infrastructure, analytical tools, and research environment that made this study feasible.

References

Corsi, S. R., De Cicco, L. A., Lutz, M. A., & Hirsch, R. M. (2014). River chloride trends in snow-affected urban watersheds. *Science of the Total Environment*, 408(24), 6715–6724.

Bureau of Water Supply. Salinity Management Assessment. New York City Department of Environmental Protection, 2025, https://www.nyc.gov/assets/dep/downloads/pdf/water/drinking-water/salinity-management-assessment.pdf.

Gonzalez, X., Hubbs, D., Dailey, S., Marsellos, A. E., & Tsakiri, K. G. (2024). *Exploring climate change effects on sinkhole formation: Long-term temperature analysis in Long Island, NY (1948–2024).* In *Proceedings of the 31st Conference on the Geology of Long Island and Metropolitan New York* (pp. 1–6).

Hope, J., Marsellos, A. E., Tsakiri, K. G., et al. (2024). Groundwater variability and environmental forcing. Long Island Geology Conference.

Hubbs, D., Gonzalez, X., Rezki, A., Tecusan, K., & Marsellos, A. E. (2025). Wavelet analysis and forecasting approach using R. Long Island Geology Conference.

Kaushal, S. S., Likens, G. E., Pace, M. L., et al. (2018). Freshwater salinization syndrome. *PNAS*, 115(4), E574–E583.

Lockwood, J. G., et al. (2014). Road salt impacts on groundwater quality. *Environmental Science & Technology*, 48(12), 7049–7057.
Maliszka, M., Johnson, A., Darnaud, D., & Marsellos, A. E. (2019). Statistical analysis of groundwater and environmental variables. Long Island Geology Conference.
Marsellos, A. E., Tsakiri, K. G., & Kavalieros, A. (2016). Data mining for flood hazard events. Mohawk Watershed Symposium.
National Weather Service. "Climate." *NOAA*, https://www.weather.gov/wrh/Climate?wfo=okx.
National Center for Biotechnology Information. (2021). StatPearls . StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK566285/
Parag, A., Marsellos, A. E., Tsakiri, K. G., et al. (2025). Wavelet analysis of groundwater level fluctuations in Long Island. Long Island Geology Conference.
Percival, D. B., & Walden, A. T. (1993). *Spectral analysis for physical applications*. Cambridge University Press.
Plitnick, T. A., Marsellos, A. E., & Tsakiri, K. G. (2018). Time series regression for flood events. *Geosciences*, 8(9).
Plitnick, T. A., Marsellos, A. E., et al. (2019). Spectral analysis of environmental time series. Long Island Geology Conference.
Rienzo, A., Weinstein, P., Mecca, K., & Marsellos, A. E. (2019). Time series analysis of groundwater levels. Long Island Geology Conference.
Rodbarry, Lea, et al. "From Streets to Streams: The Impact of Road Salt on the Environment." *The Conservation Foundation*, 10 Dec. 2023, https://theconservationfoundation.org/from-streets-to-streams-the-impact-of-road-salt-on-the-environment.
Stroud Water Research Center. "Road Salt and Fresh Water." *Stroud Water Research Center*, 25 Mar. 2026, https://stroudcenter.org/salt/.
Torrence, C., & Compo, G. P. (1998). Wavelet analysis guide. *BAMS*, 79(1), 61–78.
Tran, J., Marsellos, A. E., & Tsakiri, K. G. (2019). ARIMA and time series decomposition. Geological Society of America.
Tsakiri, K. G., Marsellos, A. E., & Zurbenko, I. G. (2014). Water discharge prediction model. *Journal of Climatology*.
Tsakiri, K. G., Marsellos, A. E., & Kapetanakis, S. (2018). ANN flood prediction. *Water*, 10(9), 1158.
United States Environmental Protection Agency. (2013). Sole source aquifer program.

R package references (ordered by package name)

De Cicco, L. A., Hirsch, R. M., Lorenz, D., & Watkins, W. (2023). dataRetrieval: R packages for discovering and retrieving water data available from U.S. federal hydrologic web services. U.S. Geological Survey.
Sparks, A. H., & Hengl, T. (2023). GHCNr: Download and process NOAA Global Historical

Climatology Network data. R package.
Wickham, H. (2016). ggplot2: Elegant graphics for data analysis. Springer.
Dunnington, D. (2023). ggspatial: Spatial data framework for ggplot2. R package.
Zurbenko, I., Smith, J., & Neagu, R. (2019). kza: Kolmogorov–Zurbenko adaptive filters. R package.
Grolemund, G., & Wickham, H. (2011). Dates and times made easy with lubridate. Journal of Statistical Software, 40(3), 1–25.
Henry, L., & Wickham, H. (2023). purrr: Functional programming tools. R package.
Pebesma, E. (2018). Simple Features for R: Standardized support for spatial vector data. The R Journal, 10(1), 439–446.
Walker, K. (2023). tigris: Load census TIGER/Line shapefiles. R package.
Wickham, H., et al. (2019). Welcome to the tidyverse. Journal of Open Source Software, 4(43), 1686.
Pohlert, T. (2020). trend: Non-parametric trend tests and change-point detection. R package.
Urban Institute. (2023). USAboundaries: Historical and contemporary boundaries of the United States. R package.
Garnier, S. (2023). viridis: Default color maps from matplotlib. R package.
Roesch, A., & Schmidbauer, H. (2018). WaveletComp: Computational wavelet analysis. R package.
Zeileis, A., & Grothendieck, G. (2005). zoo: S3 infrastructure for regular and irregular time series. Journal of Statistical Software, 14(6), 1–27.